🔷 EXPERIMENT 4 — FULL GIT WORKFLOW

✅ STEP 1: Create project folder

```
mkdir data-science-project4
cd data-science-project4
```

✅ STEP 2: Initialize Git

```
git init
```

✅ STEP 3: Create branch GIT-EXAM-1

```
git checkout -b GIT-EXAM-1
```

🔷 STEP 4: Create program2.py (Palindrome)

```
touch program2.py
notepad program2.py
```

✍️ Code:

```
text = input("Enter string: ")

if text == text[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

🔷 STEP 5: Create program1.py (Squares 1–10)

```
notepad program1.py
```

✍️ Code:

```
for i in range(1, 11):
    print(i, "square =", i*i)
```

🔷 STEP 6: Commit code

```
git add .
git commit -m "Added palindrome and squares programs"
```

🔷 STEP 7: Connect GitHub repo

```
git remote add origin https://github.com/your-username/data-science-project4.git
```

🔷 STEP 8: Push to GitHub

```
git push -u origin GIT-EXAM-1
```

🔷 STEP 9: Create IGNORE directory

```
mkdir IGNORE
cd IGNORE
```

Create files:

```
touch test1.txt
touch test2.txt
```

Go back:

```
cd ..
```

🔷 STEP 10: Create .gitignore file

```
notepad .gitignore
```

Add inside .gitignore:

```
IGNORE/
```

✔ This tells Git:

Do NOT track IGNORE folder

🔷 STEP 11: Check Git status

git status

✔ IGNORE folder should appear as ignored

🔹 STEP 12: Commit .gitignore

git add .gitignore
git commit -m "Added gitignore to ignore IGNORE folder"

🔹 STEP 13: Push changes

git push origin GIT-EXAM-1

🔹 STEP 14: Verify behavior

Run:

git status

✔ You should see:

- program files tracked
- IGNORE folder NOT tracked